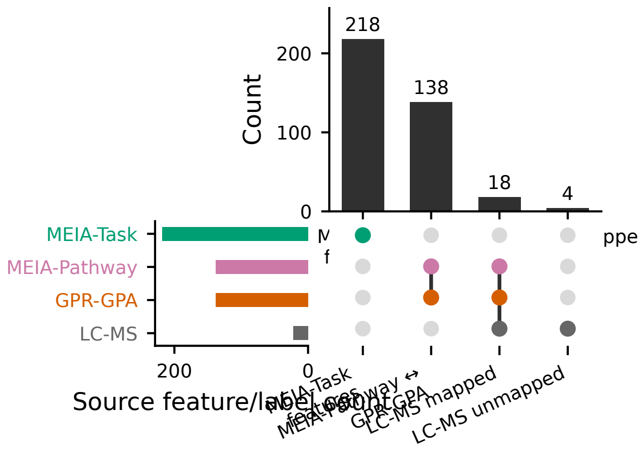
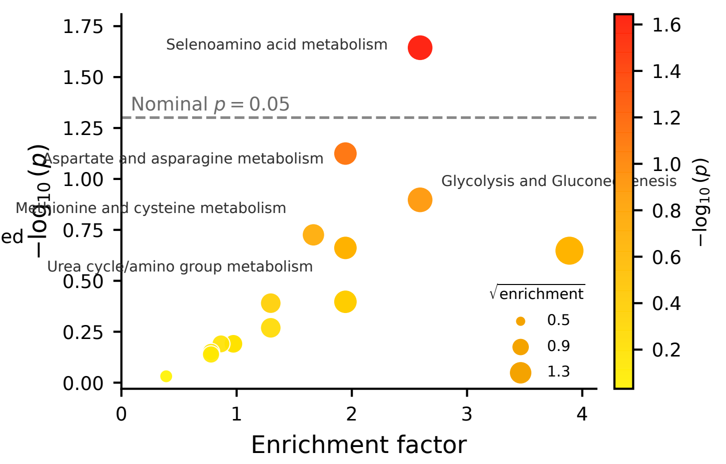
A Treatment four-set metabolic overlap

B Treatment-associated pathway enrichment

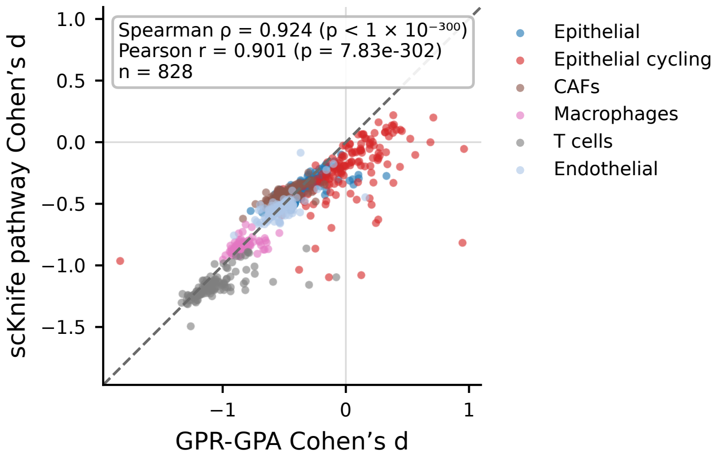
Blank-corrected LC-MS pathways: 18/22 mapped; 4 unmapped. MEIA = integrated Mu (218 tasks + 138 pathways). 22 = 356 features; top five labeled; 3 pass pathway BH-FDR < 0.05.





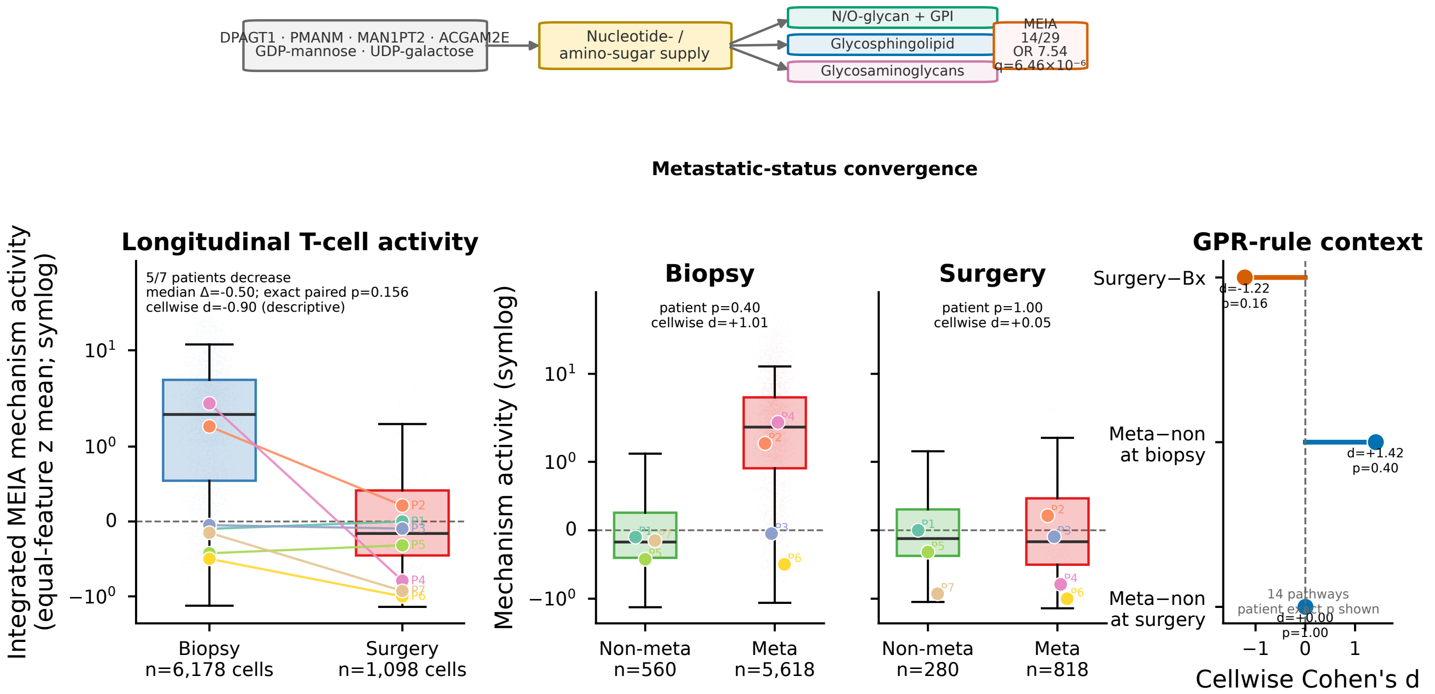
C GPR-GPA versus scKnife pathway activity

Surgery − Biopsy



D T-cell glycoconjugate activity: longitudinal decrease and postoperative convergence

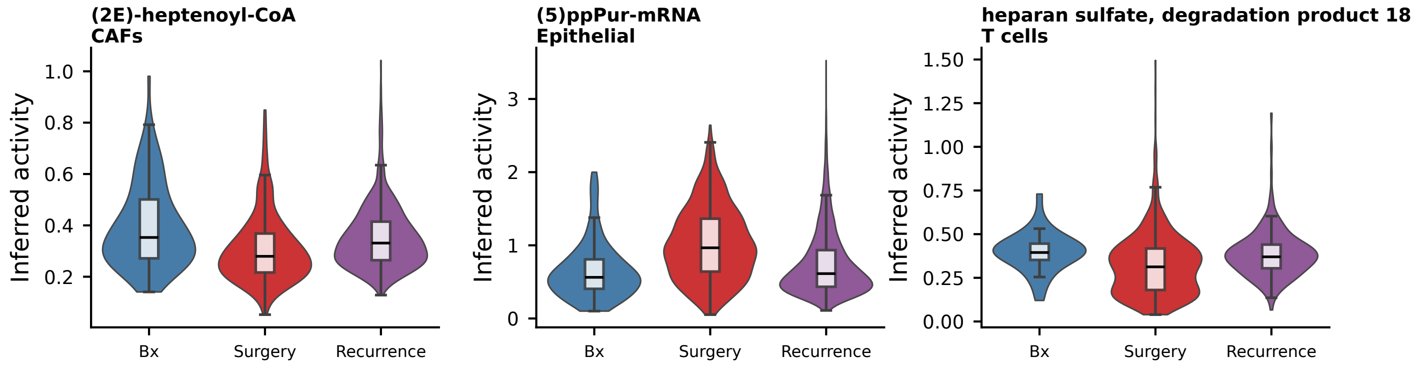
Boxes/points: all T cells; large labelled dots: P1–P7 medians; patient-level exact tests



Cell distributions and Cohen's d are descriptive because cells are nested and unequal across patients; GPR-GPA is contextual, not an independent discovery.

E Baseline-restoration candidates

Representative strict Tier-3 candidate per eligible cell type.



Tier 3 requires treatment response, recurrence rebound, and recurrence near baseline; cells are the statistical unit

F L-threonine across modalities

Paired blank-corrected LC–MS response and biopsy–surgery–recurrence scKnife activity; LC–MS q=0.0535 (exploratory).